STACKS, DISCRIMINANTS: LEMMA-DUALIZING-ASSOCIATED-PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 508-513 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 515-552, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_discriminant.json (pb=1666, stmt 508-513, proof 515-552).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05BZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 552–552. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT3: sources/stacks-discriminant/pinned-original-source.tex, source lines 450–454. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a quasi-finite homomorphism of Noetherian rings. Then $Ass_B(\omega_{B/A})$ is the set of primes of B lying over associated primes of A.*

Proof. Choose a factorization $A \to B' \to B$ with $A \to B'$ finite and $B' \to B$ inducing an open immersion on spectra. As $\omega_{B/A} = \omega_{B'/A} \otimes_{B'} B$ it suffices to prove the statement for $\omega_{B'/A}$. Thus we may assume $A \to B$ is finite.

Assume $\mathfrak{p} \in \mathrm{Ass}(A)$ and $\mathfrak{q}$ is a prime of B lying over $\mathfrak{p}$. Let $x \in A$ be an element whose annihilator is $\mathfrak{p}$. Choose a nonzero $\kappa(\mathfrak{p})$ linear map $\lambda : \kappa(\mathfrak{q}) \to \kappa(\mathfrak{p})$. Since $A/\mathfrak{p} \subset B/\mathfrak{q}$ is a finite extension of rings, there is an $f \in A$, $f \notin \mathfrak{p}$ such that $f\lambda$ maps $B/\mathfrak{q}$ into $A/\mathfrak{p}$. Hence we obtain a nonzero A-linear map

$$B \to B/\mathfrak{q} \to A/\mathfrak{p} \to A, \quad b \mapsto f\lambda(b)x$$

An easy computation shows that this element of $\omega_{B/A}$ has annihilator $\mathfrak{q}$, whence $\mathfrak{q} \in \mathrm{Ass}(\omega_{B/A})$.

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Conversely, suppose that $\mathfrak{q} \subset B$ is a prime ideal lying over a prime $\mathfrak{p} \subset A$ which is not an associated prime of A. We have to show that $\mathfrak{q} \notin \mathrm{Ass}_B(\omega_{B/A})$. After replacing A by $A_\mathfrak{p}$ and B by $B_\mathfrak{p}$ we may assume that $\mathfrak{p}$ is a maximal ideal of A. This is allowed by Lemma 0BT3 and Algebra, Lemma 05BZ. Then there exists an $f \in \mathfrak{m}$ which is a nonzerodivisor on A. Then f is a nonzerodivisor on $\omega_{B/A}$ and hence $\mathfrak{q}$ is not an associated prime of this module. $\square$